Databases

Assignment 1 - Paris Arrow Transit (PAT)

Assignment 1 Logical Checklist

Required Actions:

- ❏ Carefully checked the Marking Rubric in the assignment document so you are aware of the mark allocation
- ❏ Normalisation
  - ❏ Normalised two forms (Vehicle Assets and Driver Job Sheet [two samples are provided for the job sheet but only normalise each type of form once]). Normalised each from UNF to 3NF, showing all stages and dependencies at each step (i.e. partial dependencies in 1NF, transitive dependencies in 2NF, full dependencies in 3NF)
  - ❏ included all candidate keys for each relation at 1NF
  - ❏ Included all attributes shown on the forms
  - ❏ Did *not* add any surrogate keys in normalisation
  - ❏ Clearly indicated PK's using underline
  - ❏ Completed attribute synthesis
- ❏ Logical Model
  - ❏ Followed the unit logical model notation/requirements:
    - ❏ Used Crow's foot/Information Engineering notation for the logical model
    - ❏ Does *not* show data types and sizes
    - ❏ Does show legend on the logical model
    - ❏ Does show a label for each relationship
    - ❏ Added common prefix for all attributes in each relation (e.g. emp_no, emp_name)
    - ❏ Added comments for all attributes (used comments in RDBMS)
  - ❏ Added at least one surrogate key to the logical model and added an explanation of why chosen to the assumptions document
  - ❏ No relation with more than two attributes in the key remains (if more than two, a surrogate must be added)
  - ❏ Included all required check clauses and lookup tables
  - ❏ Included FULL normalisation results (final 3NF) in the logical model - all relations and attributes
  - ❏ Checked that the relation/attribute names in the normalisation document are identical to the logical model relation/attribute names on your model
  - ❏ For each relation, all attributes are shown on the diagram (no downward pointing diamond displayed to show some attributes are hidden)
  - ❏ Made sure there are no data anomalies in the final logical model (ie. all relations are in 3NF)
- ❏ Schema File
  - ❏ Schema file has extension .sql
  - ❏ Included DROP table commands at the head of the schema file
  - ❏ Has not been edited other than to add a header (student details) and SPOOL/ECHO commands
  - ❏ Captured run of schema file via SPOOL/ECHO commands

Required files have been pushed to the FIT GitLab server (at least 9 pushes of the model):

- ❏ pat_normalisation source file (.docx in MS Teams) and final pdf on GitLab
- ❏ pat_logical.pdf (check this is your final model and produced via File - Print Diagram - To PDF File from within Data Modeller, do not use screen capture)
- ❏ pat_oraclemodel project folder, which includes:
    - ❏ the .dmd file,
    - ❏ and the folder of the same name

  You must push this project folder while you are drawing the model with the Oracle Data Modeler
- ❏ pat_schema.sql
- ❏ pat_schema_output.txt
- ❏ pat_assumptions source file (.docx in MS Teams) and final pdf on GitLab
- ❏ pat_####_diary source file (.docx in MS Teams) and final pdf on GitLab (#### replaced by group number)

Finally, 7 individual files have been submitted to Moodle:

- ❏ pat_normalisation.pdf
- ❏ pat_logical.pdf
- ❏ pat_oraclemodel.zip, which includes:
    - ❏ the .dmd file,
    - ❏ and the model folder.

  You **MUST ensure that the zip archive of the model is tested for completeness, i.e. includes both logical and relational models** (unzipped and opened in a new location - see video: Preparing Files for Submission in ed)
- ❏ pat_schema.sql
- ❏ pat_schema_output.txt
- ❏ pat_assumptions.pdf
- ❏ pat_####_diary.pdf (#### replaced by group number)